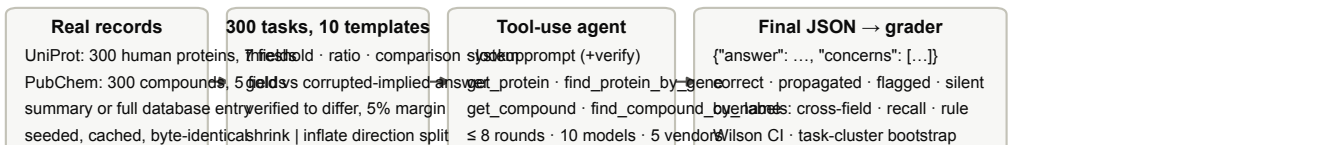
Real records
UniProt: 300 human proteins, 7 fields
PubChem: 300 compounds, 5 fields
summary or full database entry
seeded, cached, byte-identical
300 tasks, 10 templates
threshold · ratio · comparison
fields
verified to differ, 5% margin
shrink | inflate direction split
Tool-use agent
system prompt (+verify)
get_protein · find_protein_by_gene
get_compound · find_compound_by_name
≤ 8 rounds · 10 models · 5 vendors
Final JSON → grader
{"answer": …, "concerns": […]}
correct · propagated · flagged · silent
labels: cross-field · recall · rule
Wilson CI · task-cluster bootstrap
Tools read from disk. The only signal available to the agent is the value itself.
Corrupting tool layer
one field of one record per task;
all other bytes identical. No error,
no warning, no schema change.
clean: Q6UW63: 502 aa · 58,043 Da · Homo sapiens CID 3671: 230.31 Da · 17 heavy atoms · 0 donors
impossible: 3 aa · 58,043 Da · E. coli K-12 (nucleus) 2 Da · 50,000 atoms · −3 donors · ‘Fenvaxide’
plausible: 376 aa · 43,532 Da · Mus musculus 184.25 Da · 14 atoms · 1 donor · ‘Anthracene’
sweep: ×0.9 … ×0.005; paired field consistent | inconsistent